

Objective

The goal was to analyze the Windows forensic collection, recover Vera's stored credentials, decrypt protected browser data, unlock the hidden container, and retrieve the flag from the PDF.

I used impacket-secretsdump against the SYSTEM and SECURITY hives. This revealed the clear-text AutoLogon password:

```
(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ ROOT="KAPE/C"

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ ls -lh \
  "$ROOT/Windows/System32/config/SYSTEM" \
  "$ROOT/Windows/System32/config/SECURITY" \
  "$ROOT/Windows/System32/config/SOFTWARE"
-rw-rw-rw- 1 fc0d3x fc0d3x 32K Aug  4 14:19 KAPE/C/Windows/System32/config/SECURITY
-rw-rw-rw- 1 fc0d3x fc0d3x 67M Aug  4 14:19 KAPE/C/Windows/System32/config/SOFTWARE
-rw-rw-rw- 1 fc0d3x fc0d3x 10M Aug  4 14:19 KAPE/C/Windows/System32/config/SYSTEM

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ impacket-secretsdump \
  -system "$ROOT/Windows/System32/config/SYSTEM" \
  -security "$ROOT/Windows/System32/config/SECURITY" \
  LOCAL
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[*] Target system bootKey: 0x0f6f73ce89c8cda52d06fcc5131e040f
[*] Dumping cached domain logon information (domain/username:hash)
[*] Dumping LSA Secrets
[*] DefaultPassword
(Unknown User):minivera
[*] DPAPI_SYSTEM
dpapi_machinekey:0x875427f6426f5dc4e318d1e6cfed17291295e4f7
dpapi_userkey:0xb0536fa518944b2520b5a5b9f5b513e3892224a1
[*] NL$KM
0000 2F 31 EF E4 6A A6 55 5F BB 04 9E 1D 72 FD 58 42 /1..j.U....r.XB
0010 02 56 65 9D 8D 8E FD 29 56 76 CE 2F 0E 1B 29 62 .Ve....)Vv./...)b
0020 72 5A 37 BF FC 42 1F A8 58 49 ED 4D A7 85 94 A9 rZ7..B..XI.M...
0030 3E 23 9A 8B 26 1D 5F AA 8A 09 F8 41 D7 62 92 E3 >#..8.....A.b...
NL$KM:2f31efe46aa655fbb049e1d72fd58420256659d8d8efd295676ce2f0e1b2962725a37bffc421fa85849ed4da78594a93e239a8b261d5faa8a09f841d76292e3
[*] Cleaning up...
```

I decrypted it with Vera's password, this produced Vera's decrypted DPAPI masterkey.

```
(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ python3 -m venv .venv
source .venv/bin/activate

python -m pip install python-registry
Requirement already satisfied: python-registry in ./venv/lib/python3.13/site-packages (1.3.1)
Requirement already satisfied: enum-compat in ./venv/lib/python3.13/site-packages (from python-registry) (0.0.3)
Requirement already satisfied: unicodedsv in ./venv/lib/python3.13/site-packages (from python-registry) (0.14.1)

(.venv)-(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ python -c 'PY'
from Registry import Registry

h = Registry.Registry("KAPE/C/Windows/System32/config/SOFTWARE")

print("[Winlogon]")
k = h.open(r"Microsoft\Windows NT\CurrentVersion\Winlogon")

for name in ("DefaultUserName", "DefaultDomainName", "AutoAdminLogon"):
    try:
        print(f"{name}: {k.value(name).value()}")
    except Exception:
        pass

print("\n[Profiles]")
p = h.open(r"Microsoft\Windows NT\CurrentVersion\ProfileList")

for s in p.subkeys():
    try:
        print(f"{s.name()} -> {s.value('ProfileImagePath').value()}")
    except Exception:
        pass

PY
```

```
[Winlogon]
DefaultUserName: vera
DefaultDomainName:
AutoAdminLogon: 1

[Profiles]
S-1-5-18 -> %systemroot%\system32\config\systemprofile
S-1-5-19 -> %systemroot%\ServiceProfiles\LocalService
S-1-5-20 -> %systemroot%\ServiceProfiles\NetworkService
S-1-5-21-2529683458-431225740-1723070931-1000 -> C:\Users\vera
```

```
(.venv)-(fc0d3x fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ USER="vera"
SID="S-1-5-21-2529683458-431225740-1723070931-1000"

PROFILE="$ROOT/Users/$USER"
PROTECT="$PROFILE/AppData/Roaming/Microsoft/Protect/$SID"
```

```
(.venv)-(fc0d3x fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ ls -la "$PROTECT"
total 16
drwxrwxrwx 2 fc0d3x fc0d3x 4096 Aug  4 14:19
drwxrwxrwx 3 fc0d3x fc0d3x 4096 Aug  4 14:19
-rw-rw-rw- 1 fc0d3x fc0d3x 468 Aug  4 14:19 c90719ef-5b98-474e-b934-136d606a702a
-rw-rw-rw- 1 fc0d3x fc0d3x 24 Aug  4 14:19 Preferred
```

```
(.venv)-(fc0d3x fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ deactivate
```

With the password minivera I decrypted Vera's DPAPI master key from the Protect folder. This produced the usable master key needed for the next stage.

```
(fc0d3x fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ MK="$PROTECT/c90719ef-5b98-474e-b934-136d606a702a"

impacket-dpapi masterkey \
  -file "$MK" \
  -sid "$SID" \
  -password 'minivera' |
tee work/masterkey.out
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[MASTERKEYFILE]
Version      : 2 (2)
Guid         : c90719ef-5b98-474e-b934-136d606a702a
Flags        : 5 (5)
Policy       : 0 (0)
MasterKeyLen : 000000b0 (176)
BackupKeyLen : 00000090 (144)
CredHistLen  : 00000014 (20)
DomainKeyLen : 00000000 (0)

Decrypted key with User Key (SHA1)
Decrypted key: 0x5e5715ec9b6df5a86e97902692a66d28e691f05d5bc1e04d0159cfe960e94c978c07e5004a0179d3a96df2468885a28175b0b02cc064445f116a752d2b3e9d40
```

```
(fc0d3x fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ MASTERKEY="$(sed -n 's/"/Decrypted key: //p' work/masterkey.out | tail -n 1)"

echo "$MASTERKEY"
0x5e5715ec9b6df5a86e97902692a66d28e691f05d5bc1e04d0159cfe960e94c978c07e5004a0179d3a96df2468885a28175b0b02cc064445f116a752d2b3e9d40
```

```
(fc0d3x fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ find "$PROFILE/AppData" \
  \( -name 'Local State' -o -name 'Login Data' \) \
  -type f -print 2>/dev/null
KAPE/C/Users/vera/AppData/Local/Google/Chrome For Testing/User Data/Default/Login Data
KAPE/C/Users/vera/AppData/Local/Google/Chrome For Testing/User Data/Local State
```

```
(fc0d3x fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ BROWSER="$PROFILE/AppData/Local/Google/Chrome For Testing/User Data"
LOCALSTATE="$BROWSER/Local State"
LOGINDATA="$BROWSER/Default/Login Data"
```

I extracted the encrypted key from Chrome's Local State file, stripped the DPAPI prefix, and decrypted it with the master key.

```
(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ mkdir -p work

python3 - <<'PY'
import json
import base64

path = "KAPE/C/Users/vera/AppData/Local/Google/Chrome For Testing/User Data/Local State"

with open(path, "r", encoding="utf-8") as f:
    data = json.load(f)

raw = base64.b64decode(data["os_crypt"]["encrypted_key"])

print("decoded length:", len(raw))
print("prefix:", raw[:5])

if raw.startswith(b"DPAPI"):
    raw = raw[5:]
    print("Stripped DPAPI prefix")

with open("work/chrome_dpapi_blob.bin", "wb") as f:
    f.write(raw)

print("Wrote work/chrome_dpapi_blob.bin")
PY
decoded length: 317
prefix: b'DPAPI'
Stripped DPAPI prefix
Wrote work/chrome_dpapi_blob.bin
```

This gave the browser's AES-GCM key.

```
(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ impacket-dpapi unprotect \
  -file work/chrome_dpapi_blob.bin \
  -key "$MASTERKEY"
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Successfully decrypted data
0000  20 6A 39 A0 97 13 27 EA  94 87 E4 AE A9 84 4F 5D   j9...'.....0]
0010  36 70 16 24 56 98 22 76  93 9A 71 26 46 DA 0B 02   6p.$V."v..q&F...

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ BROWSERKEY="206a39a0971327ea9487e4aea9844f5d3670162456982276939a712646da0b02"
```

Using the AES key I decrypted the entry in Chrome's Login Data database. The recovered credentials were:

```

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ python3 - <<'PY'
import sqlite3

db = "KAPE/C/Users/vera/AppData/Local/Google/Chrome For Testing/User Data/Default/Login Data"

con = sqlite3.connect(f"file:{db}?mode=ro", uri=True)
cur = con.cursor()

for origin, user, pw in cur.execute("""
    SELECT origin_url, username_value, password_value
    FROM logins
"""):
    print("URL :", origin)
    print("User:", user)
    print("Prefix:", pw[:3])
    print("Hex:", pw.hex())

con.close()
PY
URL : http://bytelotus.thm:8080/
User: VeraSecretVault
Prefix: b'v10'
Hex: 763130c88a72a64f35f63e883ea0a7f64a6870e46b0bbb469a756eda88b7e324c3e1c51015aa6fd8d65ac48961e1ea324ce1707807feb3d7

```

3 bytes = "v10"
 12 bytes = GCM nonce
 N bytes = ciphertext
 16 bytes = GCM authentication tag

```

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ python3 - <<'PY'
from Cryptodome.Cipher import AES

key = bytes.fromhex(
    "206a39a0971327ea9487e4aea9844f5d3670162456982276939a712646da0b02"
)

blob = bytes.fromhex(
    "763130c88a72a64f35f63e883ea0a7f64a6870e46b0bbb469a756eda88b7e324c3e1c51015aa6fd8d65ac48961e1ea324ce1707807feb3d7"
)

assert blob[:3] == b"v10"

nonce = blob[3:15]
ciphertext = blob[15:-16]
tag = blob[-16:]

cipher = AES.new(key, AES.MODE_GCM, nonce=nonce)
plaintext = cipher.decrypt_and_verify(ciphertext, tag)

print("Password:", plaintext.decode())
PY
Password: Wh4t1sV3raD0inG0nTh1sH0st

```

Username: VeraSecretVault
 Password: Wh4t1sV3raD0inG0nTh1sH0st

```

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ find \
  "$PROFILE/Desktop" \
  "$PROFILE/Documents" \
  "$PROFILE/Downloads" \
  -type f -print 2>/dev/null
KAPE/C/Users/vera/Documents/backup

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ file "$PROFILE/Documents/backup"
KAPE/C/Users/vera/Documents/backup: data

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ xxd -l 64 "$PROFILE/Documents/backup"
00000000: f372 f7cc d607 4b17 a8aa 8865 12af abdf  .r....K....e....
00000010: f293 9a74 72ea acbc bee5 b479 4c88 5c7f  ...tr.....yL.\.
00000020: 5f53 fc44 2988 f8fc ae98 21dd c26a 2a9b  _S.D).....!..j*.
00000030: 8c45 8f73 8ac4 1cdc 8377 bb46 9636 4807  .E.s.....w.F.6H.

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ stat "$PROFILE/Documents/backup"
  File: KAPE/C/Users/vera/Documents/backup
  Size: 104857600      Blocks: 204800    IO Block: 4096   regular file
Device: 8,1      Inode: 2625317    Links: 1
Access: (0666/-rw-rw-rw-)  Uid: ( 1000/  fc0d3x)   Gid: ( 1000/  fc0d3x)
Access: 2026-08-09 12:34:28.067006238 -0400
Modify: 2026-08-04 14:15:52.000000000 -0400
Change: 2026-08-09 12:26:18.763679043 -0400
 Birth: 2026-08-09 12:26:17.602617022 -0400

```

A high-entropy 100 MB file named backup was found in Documents.

```

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ python3 - <<'py'
from collections import Counter
from math import log2

p = "KAPE/C/Users/vera/Documents/backup"
d = open(p, "rb").read()
freq = Counter(d)

entropy = -sum(
    (n / len(d)) * log2(n / len(d))
    for n in freq.values()
)

print("size:", len(d))
print("entropy:", entropy)
print("first64:", d[:64].hex())
py
size: 104857600
entropy: 7.999998250330834
first64: f372f7ccd6074b17a8aa886512afabddf2939a7472eaacbcbee5b4794c885c7f5f53fc442988f8fcae9821ddc26a2a9b8c458f738ac41cdc8377bb4696364807

```



```
(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ cryptsetup --help | grep -i tcrypt
--tcrypt-backup          Use backup (secondary) TCRYPT header
--tcrypt-hidden          Use hidden header (hidden TCRYPT
--tcrypt-system          Device is system TCRYPT drive (with
                        tcrypt, bitlk
tcryptDump <device> - dump TCRYPT device information
open: create (plainOpen), luksOpen, loopaesOpen, tcryptOpen, bitlkOpen, fvault2Open
close: remove (plainClose), luksClose, loopaesClose, tcryptClose, bitlkClose, fvault2Close

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ sudo cryptsetup tcryptDump \
--veracrypt \
"$PROFILE/Documents/backup"
[sudo] password for fc0d3x:
Enter passphrase for KAPE/C/Users/vera/Documents/backup:
VERACRYPT header information for KAPE/C/Users/vera/Documents/backup
Version:          5
Driver req.:      1.b
Flags:            0x0
Sector size:      512 [bytes]
MK offset:        131072 [bytes]
Volume size:      104595456 [bytes]
PBKDF2 hash:      sha512
Cipher chain:     aes
Cipher mode:      xts-plain64
MK bits:          512
```

cryptsetup tcryptDump --veracrypt confirmed it was a VeraCrypt volume.

```
(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ sudo cryptsetup tcryptOpen \
--veracrypt \
"$PROFILE/Documents/backup" \
vera_backup
Enter passphrase for KAPE/C/Users/vera/Documents/backup:
Device vera_backup already exists.

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ ls -l /dev/mapper/vera_backup
lrwxrwxrwx 1 root root 7 Aug  9 13:09 /dev/mapper/vera_backup -> ../dm-0

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ sudo blkid /dev/mapper/vera_backup
/dev/mapper/vera_backup: UUID="8A64-8FDE" BLOCK_SIZE="512" TYPE="vfat"
```

```
(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ sudo mkdir -p /mnt/vera_backup

sudo mount -o ro \
/dev/mapper/vera_backup \
/mnt/vera_backup

(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ find /mnt/vera_backup -type f -print
/mnt/vera_backup/$RECYCLE_BIN/desktop.ini
/mnt/vera_backup/secret_financial_documents/important_invoice_byte_lotus.pdf
/mnt/vera_backup/secret_financial_documents/transactions_q3.csv
/mnt/vera_backup/System Volume Information/WPSettings.dat
```

```
(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ file "$PDF"
pdfinfo "$PDF"
/mnt/vera_backup/secret_financial_documents/important_invoice_byte_lotus.pdf: PDF document, version 1.7, 1 page(s) (zip deflate encoded)
Custom Metadata: no
Metadata Stream: no
Tagged: no
UserProperties: no
Suspects: no
Form: none
JavaScript: no
Pages: 1
Encrypted: no
Page size: 477 x 543 pts
Page rot: 0
File size: 26747 bytes
Optimized: no
PDF version: 1.7
```

```
(fc0d3x@fc0d3x)-[~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14]
$ sudo apt install poppler-utils
poppler-utils is already the newest version (26.01.0-5).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 875
```

Inside the mounted volume an invoice PDF contained the final flag.

The image shows a dual-pane view of a computer screen. The left pane is a terminal window with a dark background. It shows a user at the prompt `(fc0d3x@fc0d3x)~` navigating to `~/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14` and creating a directory `work/pdfpages`. Then, the user runs `xdg-open work/pdfpages/page-1.png`, which results in an error: `Created TensorFlow Lite XNNPACK delegate for CPU. [95686:95712:0809/171756.849310:ERROR:google_api/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT`. The right pane shows a web browser window with the address bar displaying `/home/fc0d3x/thm/HackerHolidays26/F/management-wants-a-word-forensics-hh-day-14`. The browser shows an "INVOICE" from "BYTE LOTUS RESORTS". The invoice details include: INVOICE NR: 2122/0900/5050, DATE: 7/19/2028, DUE DATE: 8/09/2028. The "BILL TO:" section lists "Hotel Cleaning LLC, 25 SMTD Rd, Resort Drive". The "PAYABLE TO:" section lists "Byte Lotus Resorts, 8080 Http Street, Resort Drive". Below this is a table with columns: NO, DESCRIPTION, QTY, PRICE, and TOTAL. The first row of the table contains: 1, Flag: THM!t.w4s.V3r4.A11.A10ng?!!, 1, \$100, and \$100. The rest of the table is empty.